

Python for Data Science 2

Lab 7- SUSY at LHC

Amir Farbin

- To understand the most fundamental phenomena in nature, the Physicists application of the scientific method:

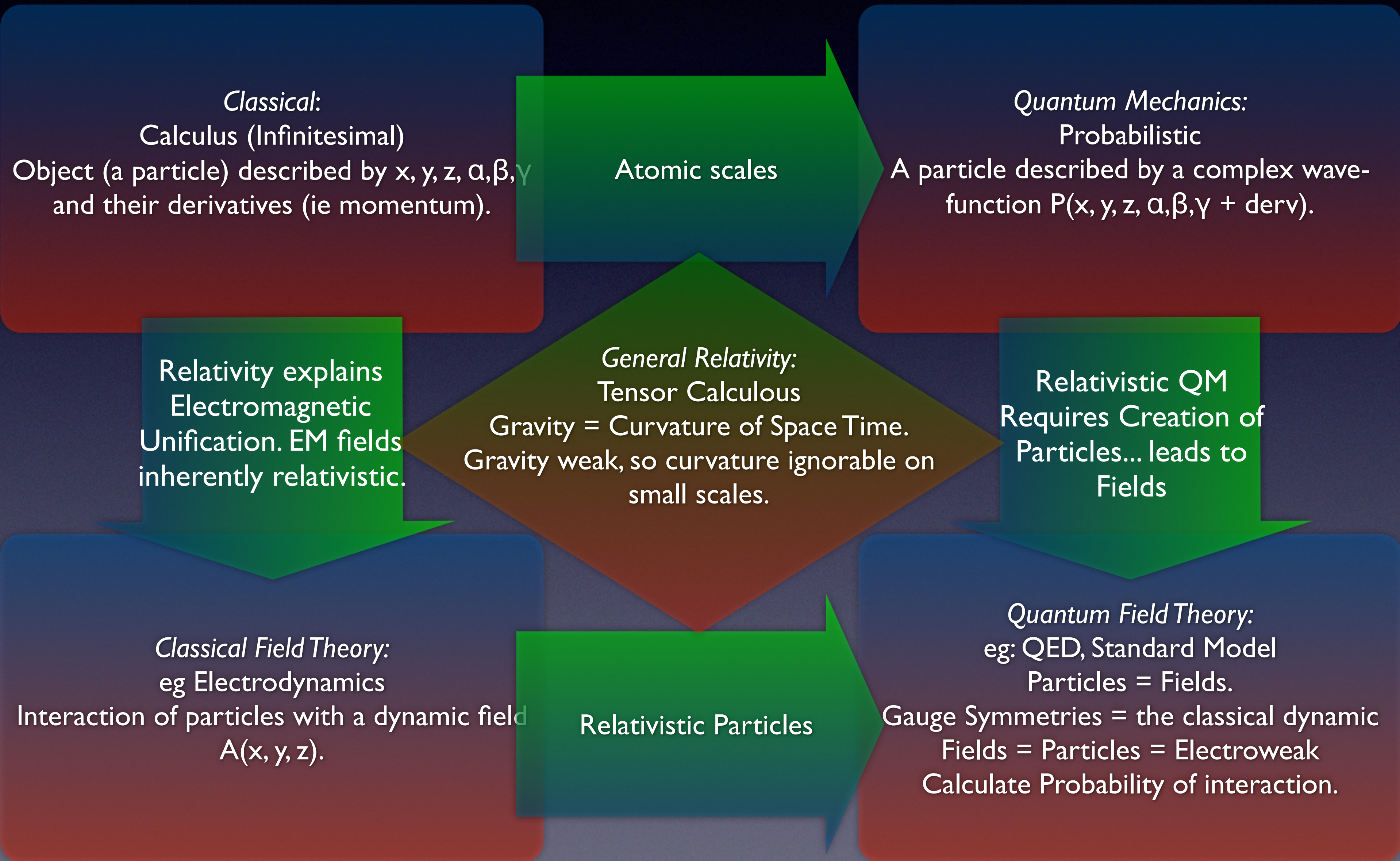
1. Build Models

- The **Standard Model of HEP** describes the building blocks of matter and their interactions.
 - SM requires the existence of a set of fundamental particles with very specific properties.
 - Tested and validated through experiments, including some of the most precise measurements ever.
 - Yet, has **failures**, e.g. no Dark Matter. Not compatible with Gravity.
 - And, has **inconsistencies**: e.g. both SM predicts the Higgs mass to be big and requires it to be small. (Known as the Hierarchy problem...)
- Models are **expressed mathematically**
 - Ideally build on a minimal set of rules (principles/laws)
 - SM Mathematical Framework: Quantum Field Theory- Way beyond this class.
 - Renormalization: encapsulating unknown (effects at high energy) in measurable parameters.
 - SM is a Model with **19 parameters**. Masses of particles. Strengths of forces. etc...
 - None are fundamental like speed of light or Planck's constant.

2. Build new model that tackle inconsistencies.

3. Create experiments (target weaknesses in the model)

Our Mathematical Models



Building the SM

Field Theory + Quantum Mechanics + Relativity = Quantum Field Theory

Ingredients

1. Leptons + Quarks

FERMIONS			matter constituents		
			spin = 1/2, 3/2, 5/2, ...		
Leptons spin = 1/2			Quarks spin = 1/2		
Flavor	Mass GeV/c ²	Electric charge	Flavor	Approx. Mass GeV/c ²	Electric charge
ν_e electron neutrino	$<1 \times 10^{-8}$	0	u up	0.003	2/3
e^- electron	0.000511	-1	d down	0.006	-1/3
ν_μ muon neutrino	<0.0002	0	c charm	1.3	2/3
μ^- muon	0.106	-1	s strange	0.1	-1/3
ν_τ tau neutrino	<0.02	0	t top	175	2/3
τ^- tau	1.7771	-1	b bottom	4.3	-1/3

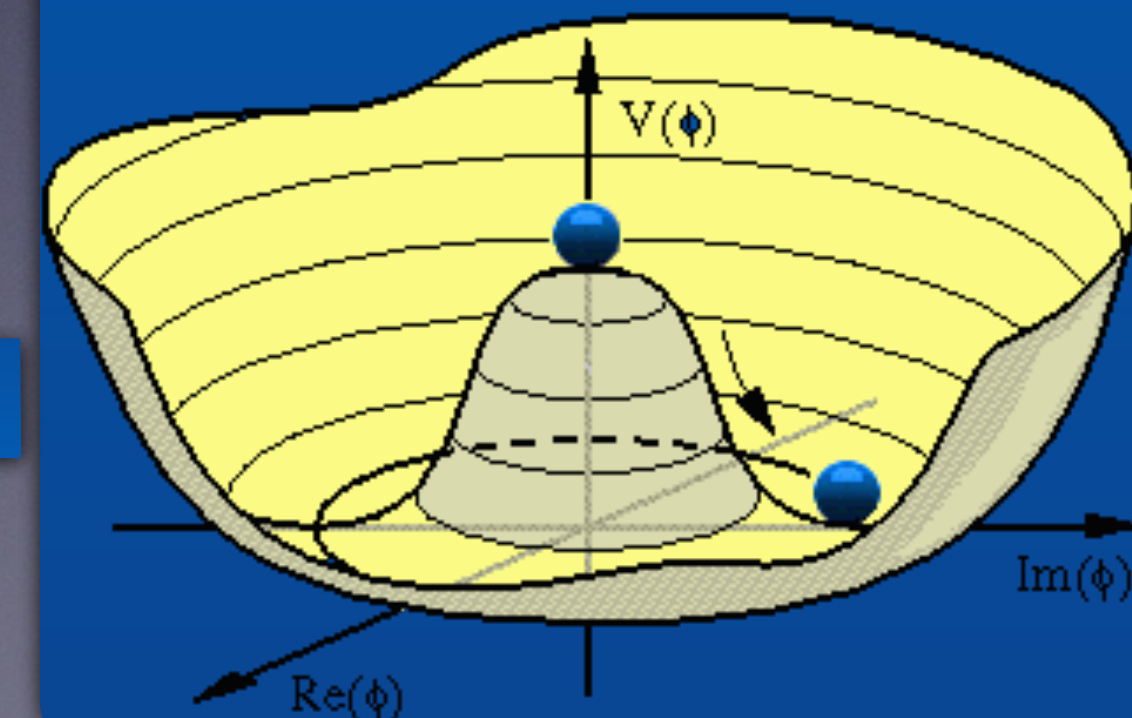
2. Three Local Gauge Symmetries

- Each Symmetry implies existence of a new set of boson (spin 1) fields
- These bosons “carry” the Forces

Unified Electroweak spin = 1			Strong (color) spin = 1		
Name	Mass GeV/c ²	Electric charge	Name	Mass GeV/c ²	Electric charge
γ photon	0	0	g gluon	0	0
W^-	80.4	-1			
W^+	80.4	+1			
Z^0	91.187	0			

3. Higgs Mechanism

- Every particle interacts with a scalar (spin 0) field
- This field has non-zero Vacuum Expectation Value
 - Breaks symmetry between E&M and Weak Forces
 - Gives masses to all particles w/o breaking Gauge Symmetries

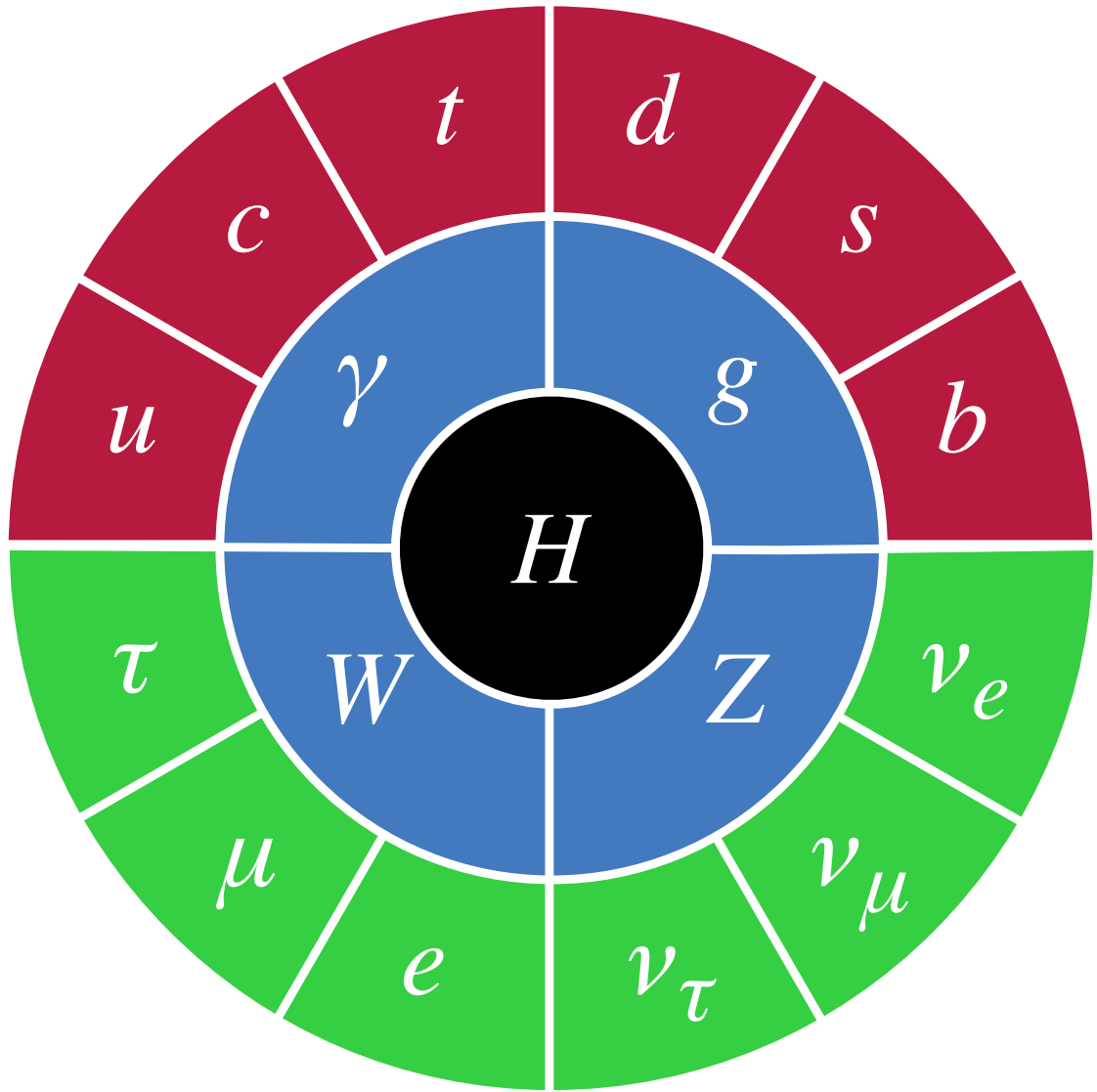


Standard Model

Agreement with all experimental results so far

PARTICLE PHYSICS: 19 PARAMETERS

$$\begin{aligned} \mathcal{L}_{SM} = & \underbrace{\frac{1}{4}\mathbf{W}_{\mu\nu} \cdot \mathbf{W}^{\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}G_{\mu\nu}^a G_a^{\mu\nu}}_{\text{kinetic energies and self-interactions of the gauge bosons}} \\ & + \underbrace{\bar{L}\gamma^\mu(i\partial_\mu - \frac{1}{2}g\boldsymbol{\tau} \cdot \mathbf{W}_\mu - \frac{1}{2}g'YB_\mu)L + \bar{R}\gamma^\mu(i\partial_\mu - \frac{1}{2}g'YB_\mu)R}_{\text{kinetic energies and electroweak interactions of fermions}} \\ & + \underbrace{\frac{1}{2}\left|(i\partial_\mu - \frac{1}{2}g\boldsymbol{\tau} \cdot \mathbf{W}_\mu - \frac{1}{2}g'YB_\mu)\phi\right|^2 - V(\phi)}_{W^\pm, Z, \gamma, \text{and Higgs masses and couplings}} \\ & + \underbrace{g''(\bar{q}\gamma^\mu T_a q)G_\mu^a}_{\text{interactions between quarks and gluons}} + \underbrace{(G_1\bar{L}\phi R + G_2\bar{L}\phi_c R + h.c.)}_{\text{fermion masses and couplings to Higgs}} \end{aligned}$$



Symbol	Description	Value
m_e	Electron mass	511 keV
m_μ	Muon mass	105.7 MeV
m_τ	Tau mass	1.78 GeV
m_u	Up quark mass	1.9 MeV
m_d	Down quark mass	4.4 MeV
m_s	Strange quark mass	87 MeV
m_c	Charm quark mass	1.32 GeV
m_b	Bottom quark mass	4.24 GeV
m_t	Top quark mass	172.7 GeV
θ_{12}	CKM 12-mixing angle	13.1°
θ_{23}	CKM 23-mixing angle	2.4°
θ_{13}	CKM 13-mixing angle	0.2°
δ	CKM CP-violating Phase	0.995
g_1	U(1) gauge coupling	0.357
g_2	SU(2) gauge coupling	0.652
g_3	SU(3) gauge coupling	1.221
θ_{QCD}	QCD vacuum angle	~0
v	Higgs vacuum expectation value	246 GeV
m_H	Higgs mass	125 GeV

Standard Model of
FUNDAMENTAL PARTICLES AND INTERACTIONS

The Standard Model summarizes the current knowledge in Particle Physics. It is the quantum theory that includes the theory of strong interactions (quantum chromodynamics or QCD) and the unified theory of weak and electromagnetic interactions (electroweak). Gravity is included on this chart because it is one of the fundamental interactions even though not part of the "Standard Model."

FERMIONS

matter constituents

spin = 1/2, 3/2, 5/2, ...

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μ muon	0.106	-1	s strange	0.1	-1/3
ν_τ tau neutrino	<0.02	0	b bottom	4.3	-1/3
τ tau	1.7771	-1			

Spin is the intrinsic angular momentum of a particle. It is a quantum unit of angular momentum.

Electric charges are given in units of the proton's charge. In SI units the electric charge of the proton is 1.60×10^{-19} coulombs.

The energy unit of particle physics is the electronvolt (eV), the energy gained by one electron in crossing a potential difference of one volt. Masses are given in GeV/c² (remember $E = mc^2$), where 1 GeV = 10^9 eV = 1.60×10^{-10} joule. The mass of the proton is 0.938 GeV/c² = 1.67×10^{-27} kg.

BOSONS

force carriers

spin = 0, 1, 2, ...

Unified Electroweak spin = 1			Strong (color) spin = 1		
Name	Mass GeV/c ²	Electric charge	Name	Mass GeV/c ²	Electric charge
γ photon	0	0	g gluon	0	0
W^-	80.4	-1			
W^+	80.4	+1			
Z^0	91.187	0			

Color Charge

Each quark carries one of three types of "strong charge," also called "color charge." These charges have nothing to do with the colors of visible light. There are eight possible types of color charge for gluons. Just as electrical charges interact by exchanging photons, in strong interactions color-charged particles interact by exchanging gluons. Leptons, photons, and W and Z bosons have no strong interactions and hence no color charge.

Quarks Confined in Mesons and Baryons

One cannot isolate quarks and gluons; they are confined in color-neutral particles called hadrons. This confinement (binding) results from multiple exchanges of gluons among the quarks. As a result, quarks are never found in isolation. An example of this is shown below. Two additional quark-antiquark pairs (see figure below). The quarks and antiquarks then combine into hadrons; these are the particles seen to emerge. Two types of hadrons have been observed in nature: mesons $q\bar{q}$ and baryons qqq .

Residual Strong Interaction

The strong binding of color-neutral protons and neutrons to form nuclei is due to residual strong interactions between their color-charged constituents. It is similar to the residual electrical interaction that binds electrically neutral atoms to form molecules. It can also be viewed as the exchange of mesons between the hadrons.

Structure within the Atom

Quark

Size $< 10^{-19}$ m

Electron

Size $\sim 10^{-18}$ m

Proton

Size $\sim 10^{-15}$ m

Atom

Size $\sim 10^{-10}$ m

Size of quark and electron within the proton is smaller than the proton size and the entire atom would be about 10 km across.

Interaction	Gravitational	Weak	Electromagnetic	Strong
		(Electroweak)		
Acts on:	Mass – Energy	Flavor	Electric Charge	Color Charge
Particles experiencing:	All	Quarks, Leptons	Electrically charged	Quarks, Gluons
Mediating particles:	Graviton	W^+, W^-, Z^0	γ	g
Strengths relative to electromagnetic 10^{-16} m for two 11 quarks at:	10^{-41}	0.8	1	25
	10^{-41}	10^{-4}	1	60
	10^{-36}	10^{-7}	1	Not applicable to hadrons
				20

Mesons $q\bar{q}$					
Mesons are bosonic hadrons. There are about 140 types of mesons.					
Symbol	Name	Quark content	Electric charge	Mass GeV/c ²	Spin
π^+	pion	$u\bar{d}$	+1	0.140	0
π^-	anti-pion	$\bar{u}d$	-1	0.140	0
ρ^+	rho	$u\bar{d}$	+1	0.770	1
B^0	B-zero	$d\bar{b}$	0	5.279	0
η_c	eta-c	$c\bar{c}$	0	2.980	0

Matter and Antimatter

For every particle type there is a corresponding antiparticle, represented by a bar over the particle symbol (unless the charge is shown). Particle and antiparticle have identical mass and spin but opposite charges. Some electrically neutral bosons (e.g., Z^0 , γ , and $\eta_c = c\bar{c}$, but not $K^0 = d\bar{s}$) are their own antiparticles.

Figures

These diagrams are an artist's conception of physical processes. They are not exact and have no meaningful scale. Green shaded areas represent the cloud of gluons or the gluon field, and red lines the quark paths.

$n \rightarrow p e^- \bar{\nu}_e$

A neutron decays to a proton, an electron, and an antineutrino via a virtual (mediating) W boson. This is neutron β decay.

$e^+ e^- \rightarrow B^0 \bar{B}^0$

An electron and positron (antielectron) colliding at high energy can annihilate to produce B^0 and $\bar{B}^0$ mesons via a virtual Z boson or a virtual photon.

$p p \rightarrow Z^0 Z^0 + \text{assorted hadrons}$

Two protons colliding at high energy can produce various hadrons plus very high mass particles such as Z bosons. Events such as this one are rare but can yield vital clues to the structure of matter.

The Particle Adventure

For more information, please feature The Particle Adventure at www.cpepweb.org

This chart has been made possible by the generous support of:

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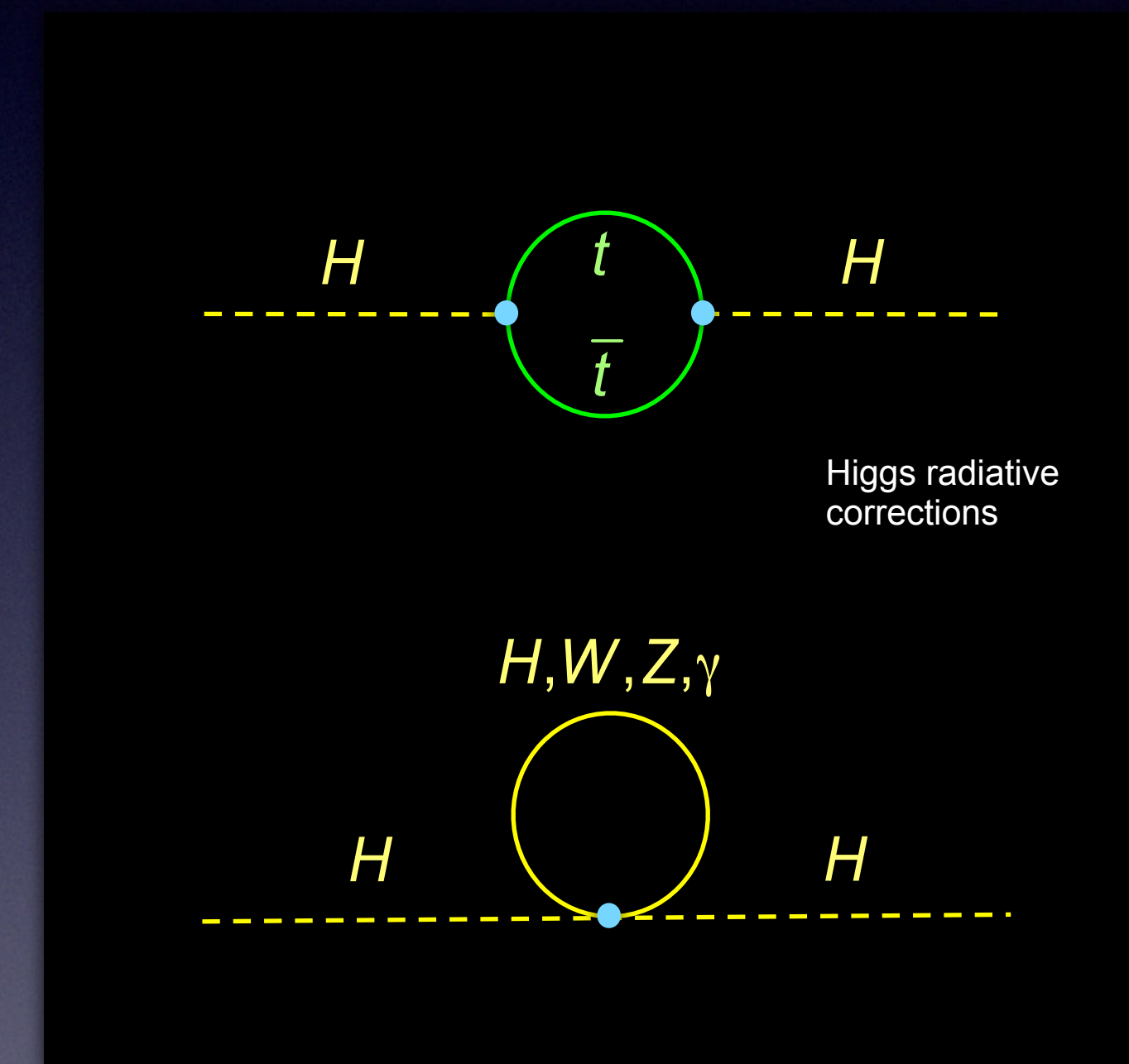
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- Why Look Beyond the Standard Model?
- Takes 19 parameters (eg Masses)... Why these values?
- Gravity not included! Why gravity is so much weaker than everything else?
- Misses a lot of the Universe: No Dark matter candidate. Can't explain Dark Energy.
- Doesn't have enough asymmetry between matter/anti-matter to explain why we exist!
- Higgs Mass Hierarchy Problem / Fine-tuning / Natureness

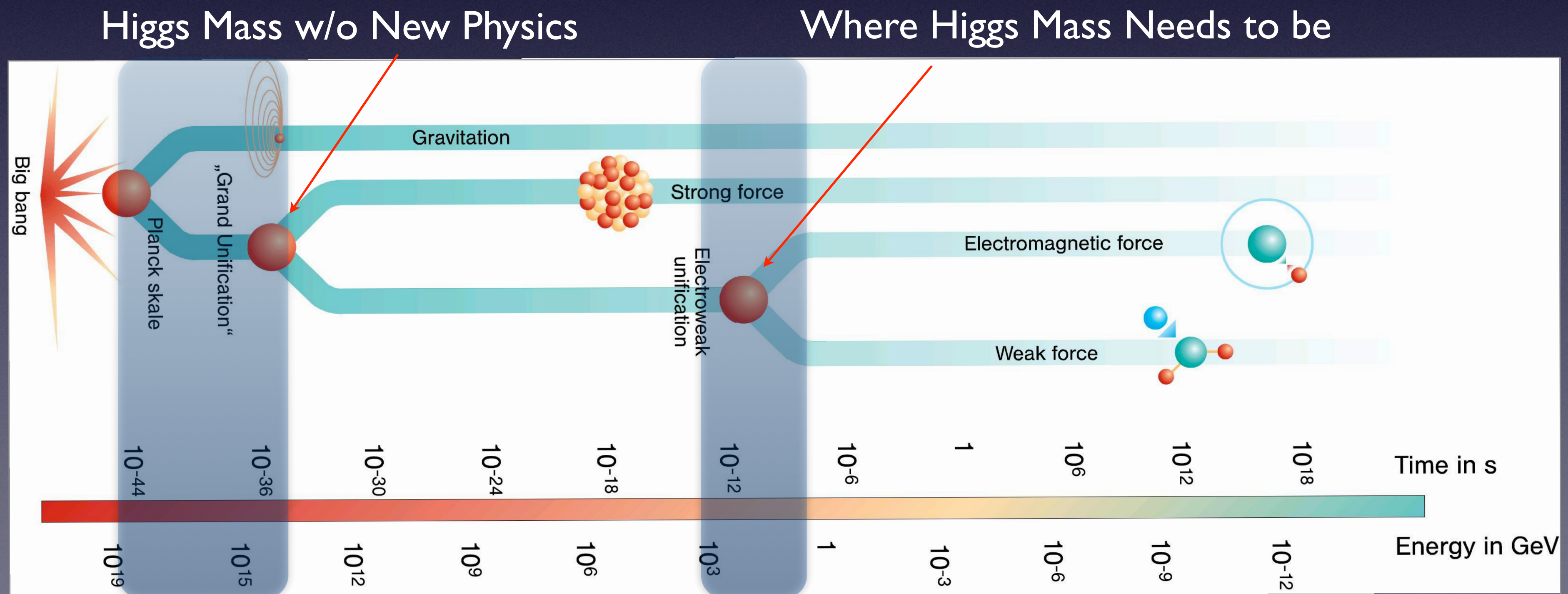
Why the Higgs Mass is a Problem?

- Think of the mass like a number that is constantly being “pushed around” by everything in the quantum vacuum
- → Every particle in nature contributes a correction to every other particles’ mass... usually small.
 - Heavy particles (like the top quark) give very large pushes
- Higgs is the only “scalar” particle in the SM, so corrections are different.
 - If physics continues to very high energies, → these pushes become enormous for the Higgs
- So naively, you would expect:
 - → The Higgs mass should be very large (set by the highest energy scale in nature)



The Puzzle

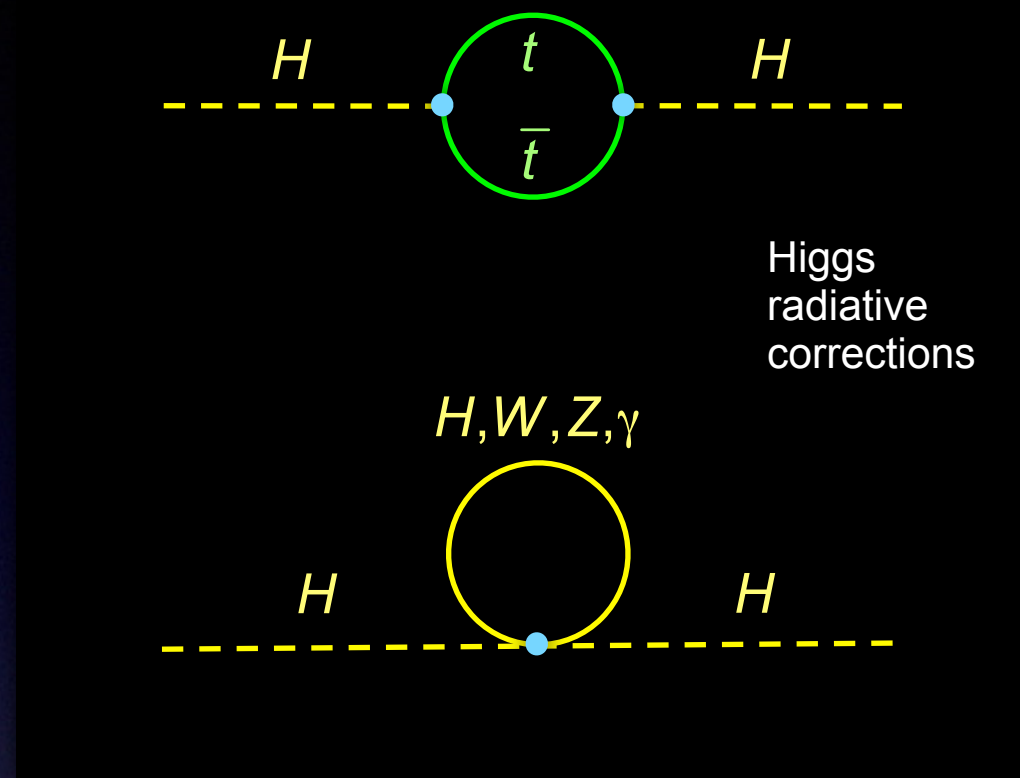
- We naively expect the Higgs to be very very Heavy due to radiative corrections.
- But experimentally, the Higgs mass is only 125 GeV
 - → much smaller than, for example, the Planck scale
- To make this happen, nature would need:
 - → huge contributions that almost perfectly cancel each other
 - This is like adding $10^{16} - 10^{16} + 125$ and somehow always getting exactly 125
 - That kind of precise cancellation feels unnatural



Why We Expect New Physics?

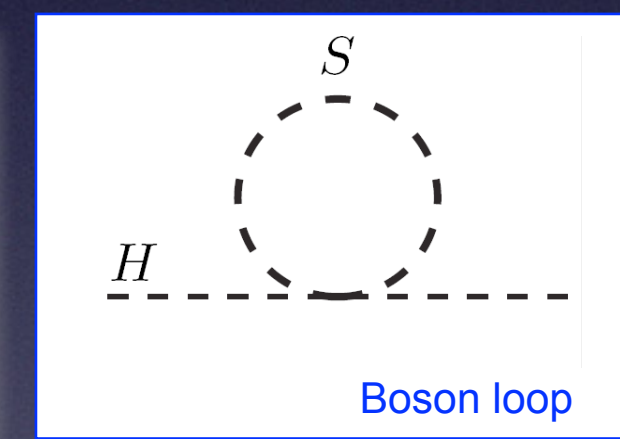
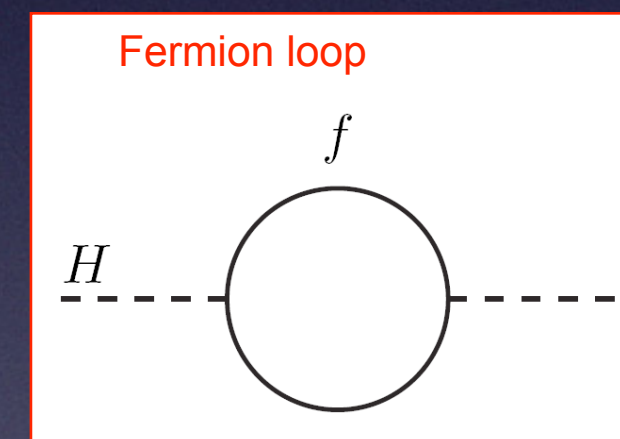
- Physicists interpret this as a sign that something is missing in the Standard Model
- The idea:
 - → New physics should automatically cancel or limit these large contributions
- Examples:
 - New particles that cancel quantum effects → Supersymmetry
 - Higgs not fundamental (composite) → Technicolor
 - Extra dimensions changing behavior at high energy → Large Extra Dimensions
- All predict new particles... usually partners to the Standard Model particles
- Some of the theories solve multiple problems:
 - Higgs Mass Fine tuning, Dark Matter, Gauge [Force] Unification

SUSY and Hierarchy Problem



- Recall the mass of the Higgs is M_{GUT} or M_{PL} , because of “infinite” radiative corrections (ie loop diagrams) that are quadratic in cut-off Λ .

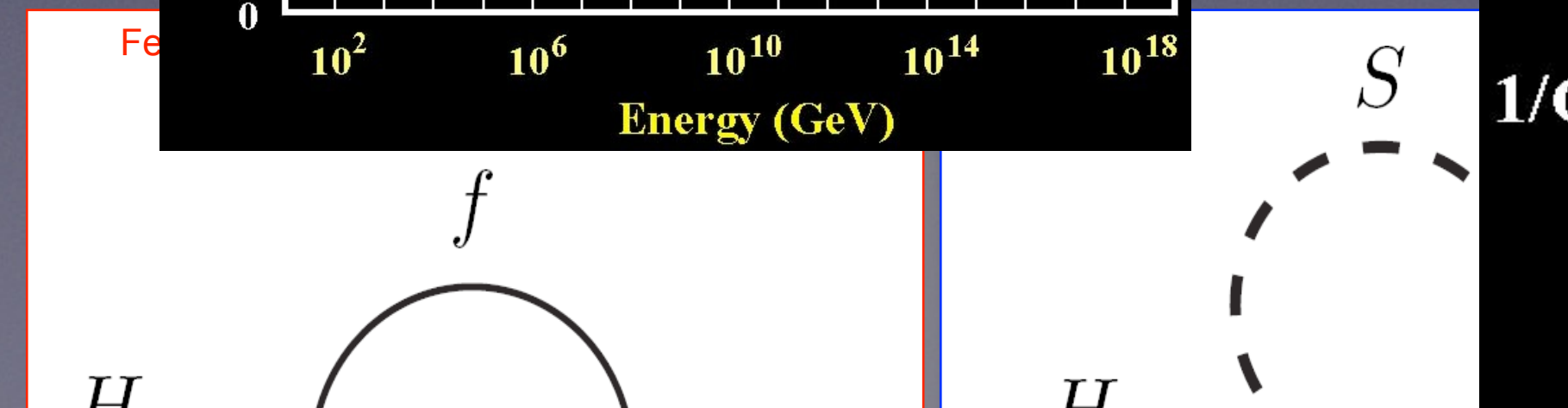
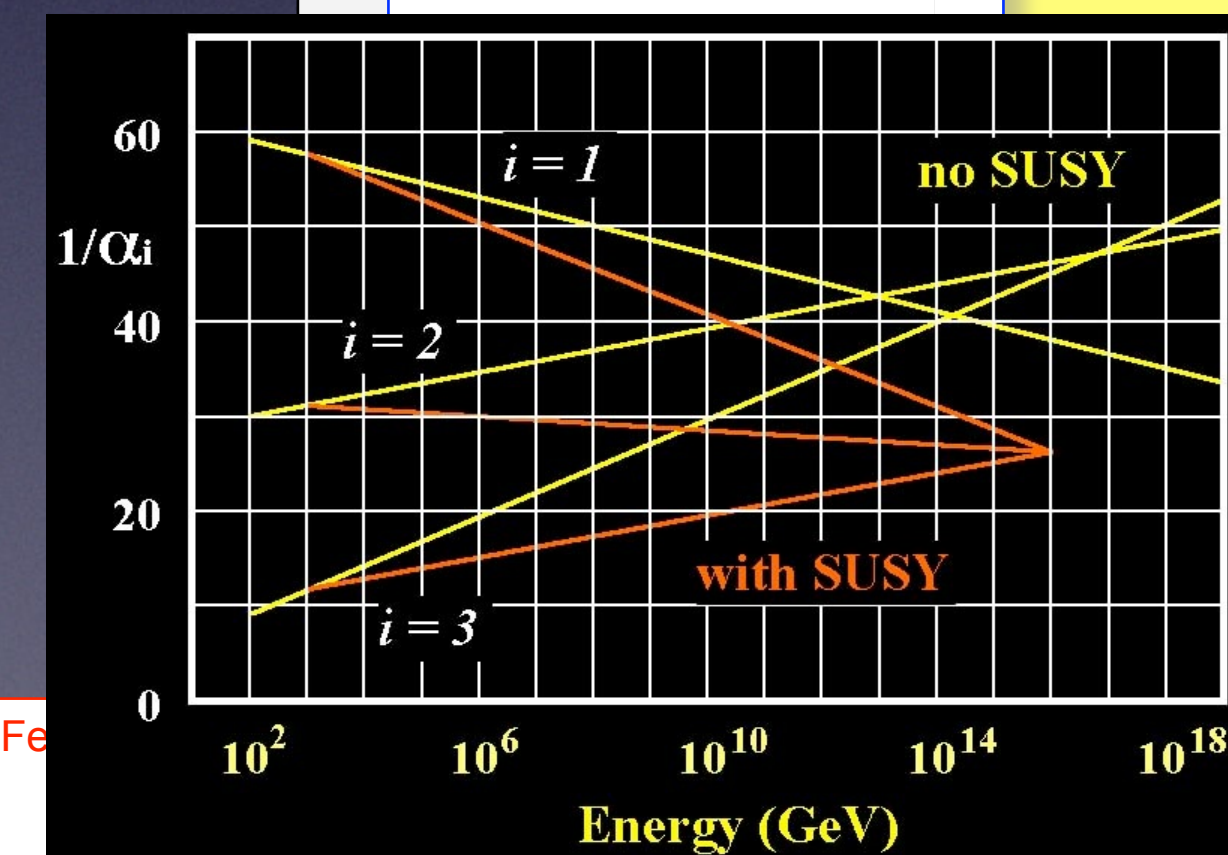
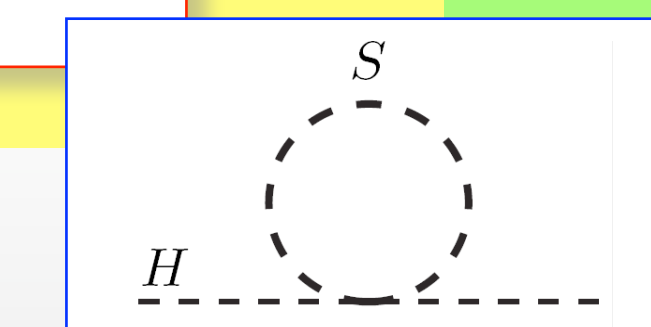
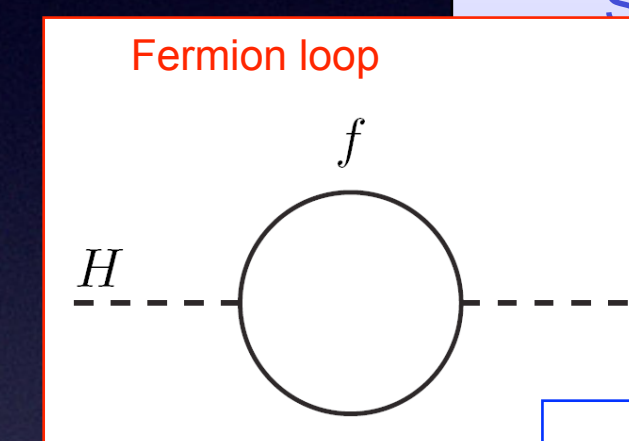
- What if for every such diagram, there existed another diagram, with opposite sign?
- Bosons/Fermions come with different signs
- A new space-time symmetry (extension of Poincare Group) results in a new boson for every fermion and visa-versa
- particle $\rightarrow$ sparticle, gauge boson $\rightarrow$ gaugino
- No scalar electron partner $\Rightarrow$ SUSY broken
- Still keeps the Higgs mass low



SUSY Motivation

- Aesthetic: new space-time symmetry
 - Leads to new partners for every SM particle.
 - Removes quadratic divergences (Higgs mass).
- ➔ Resolves Hierarchy problem
- Gauge unification
- Has Graviton
- Dark matter Candidate: The **L**ightest **S**USY **P**article can be a heavy stable neutral particle

Spin 0	Spin 1/2	Spin 1	Spin 3/2	Spin 0
Higgs	Higgsino		Gravitino	Graviton
sLepton	Lepton			
sQuark	Quark			
	Gluino	Gluon		
	Photino	Photon		
	Zino	Z		
	Wino	W		

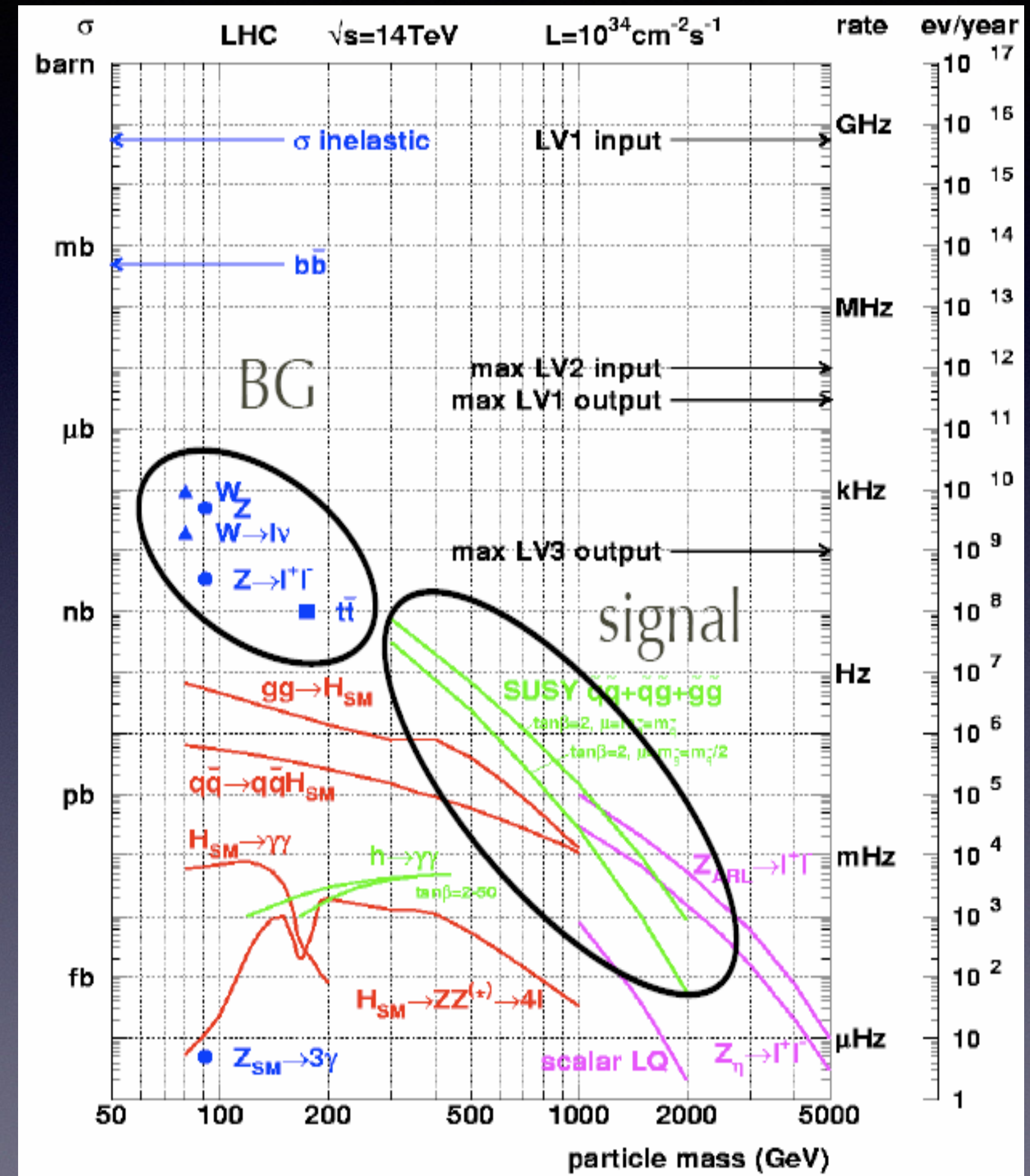


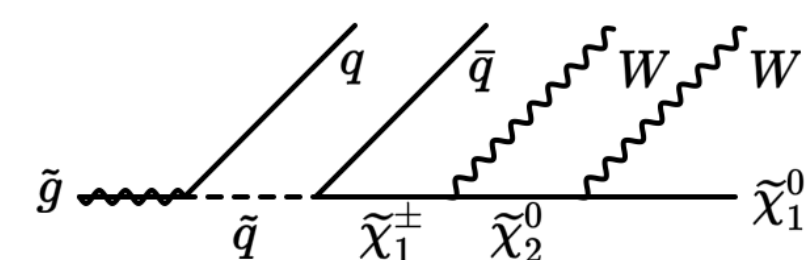
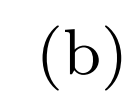
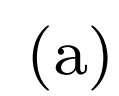
Looking for SUSY

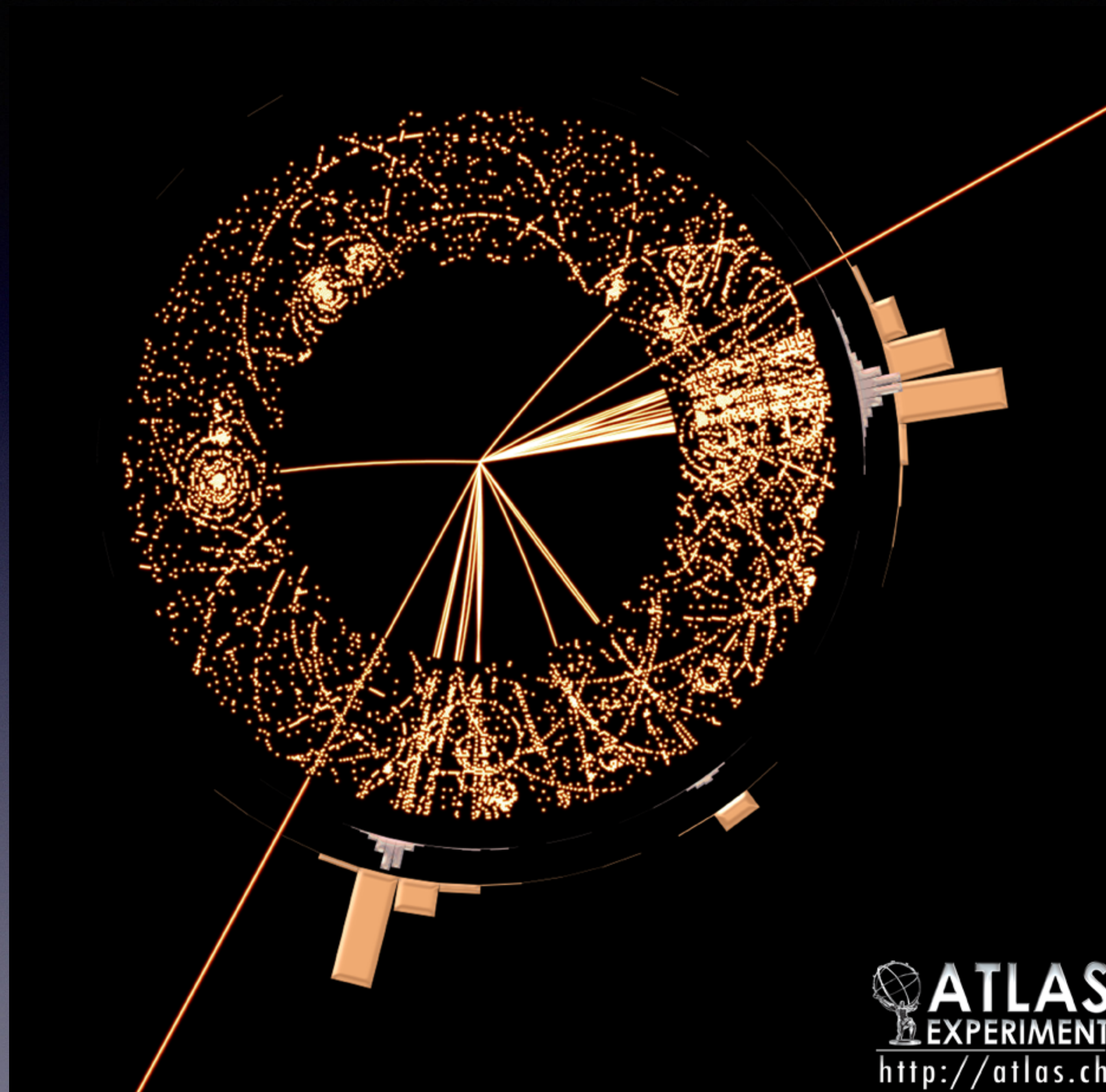


Signal vs Background

- 40 Million collisions a second
- 50 interactions per collision
- Interesting stuff (signal) is ~ 1 in a trillion
- The rest is background

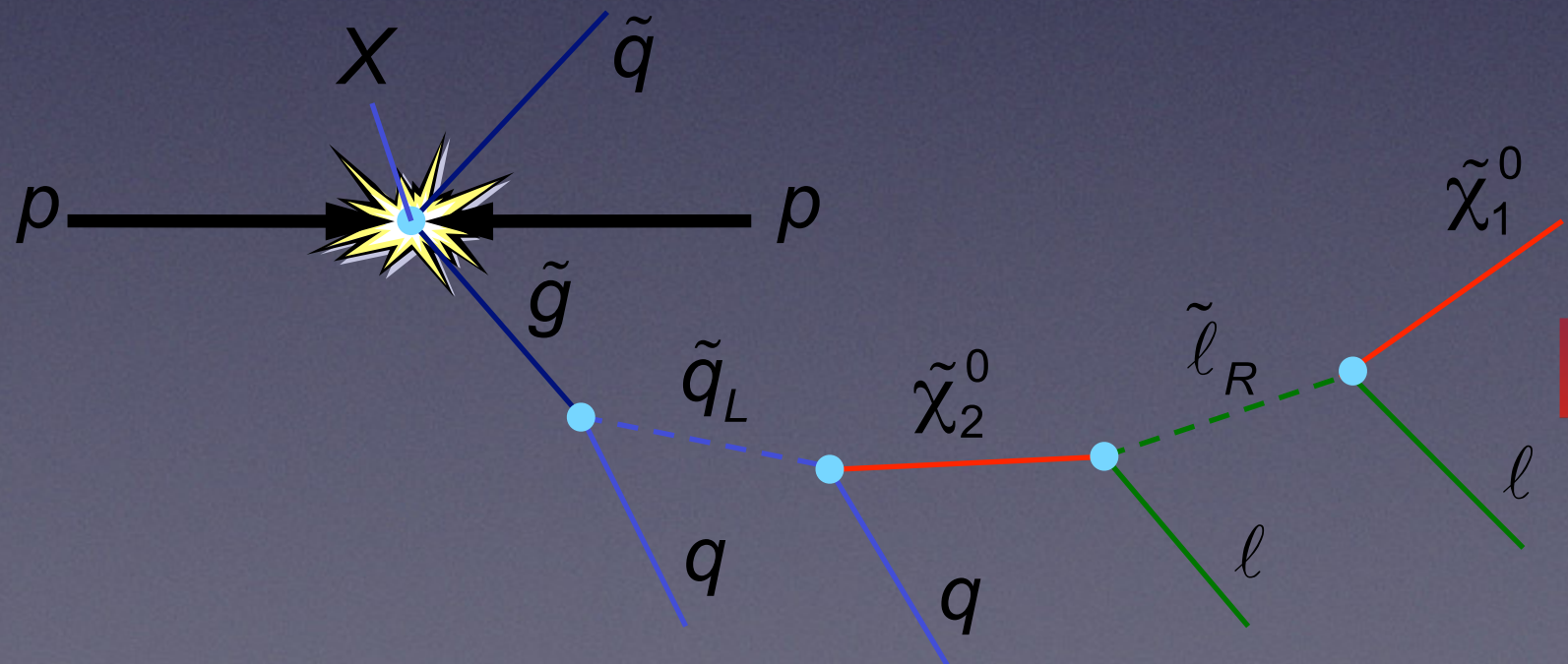
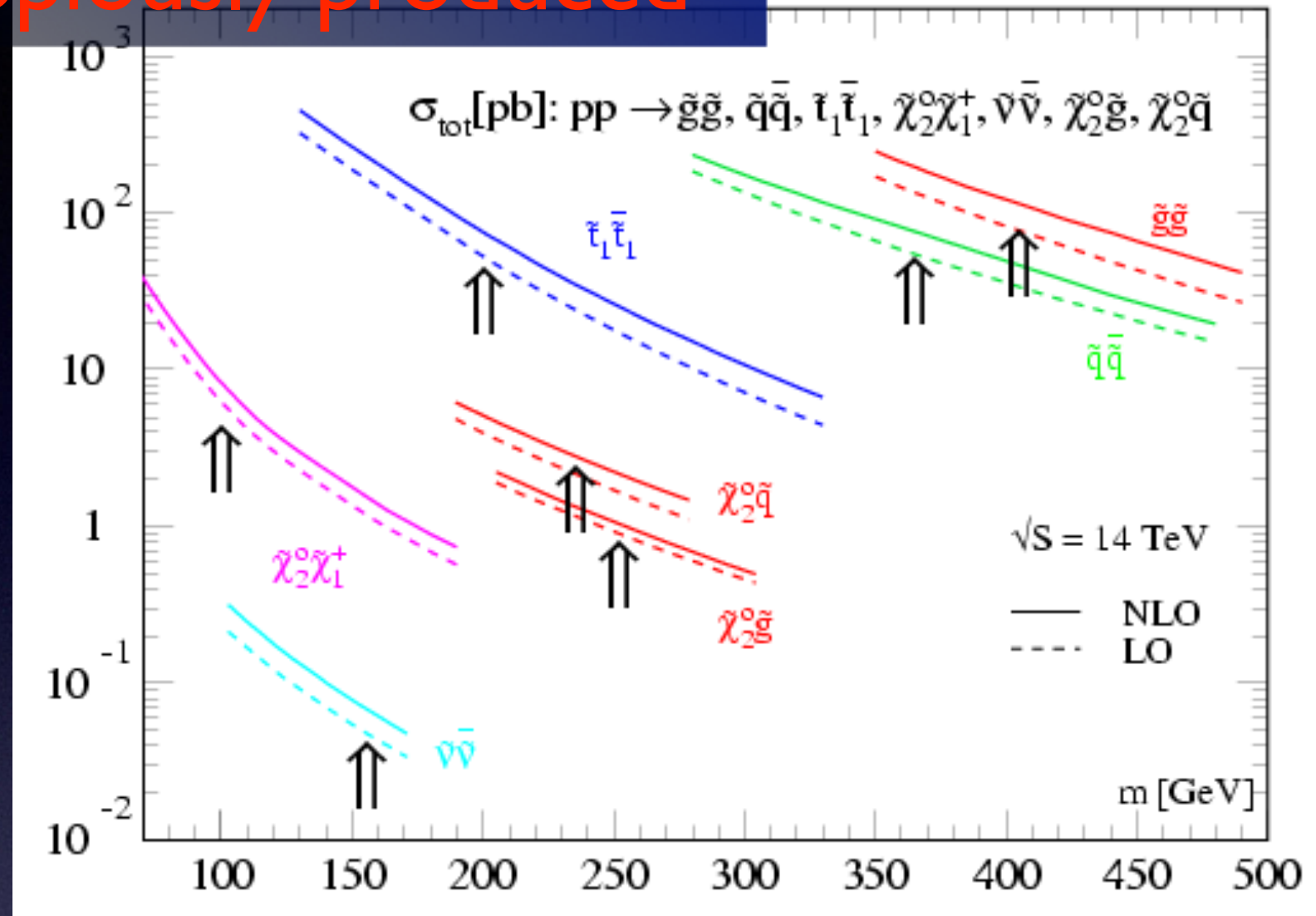






SUSY at LHC

Gluginos and squarks copiously produced*



Long Cascades

GMSB

AMSB

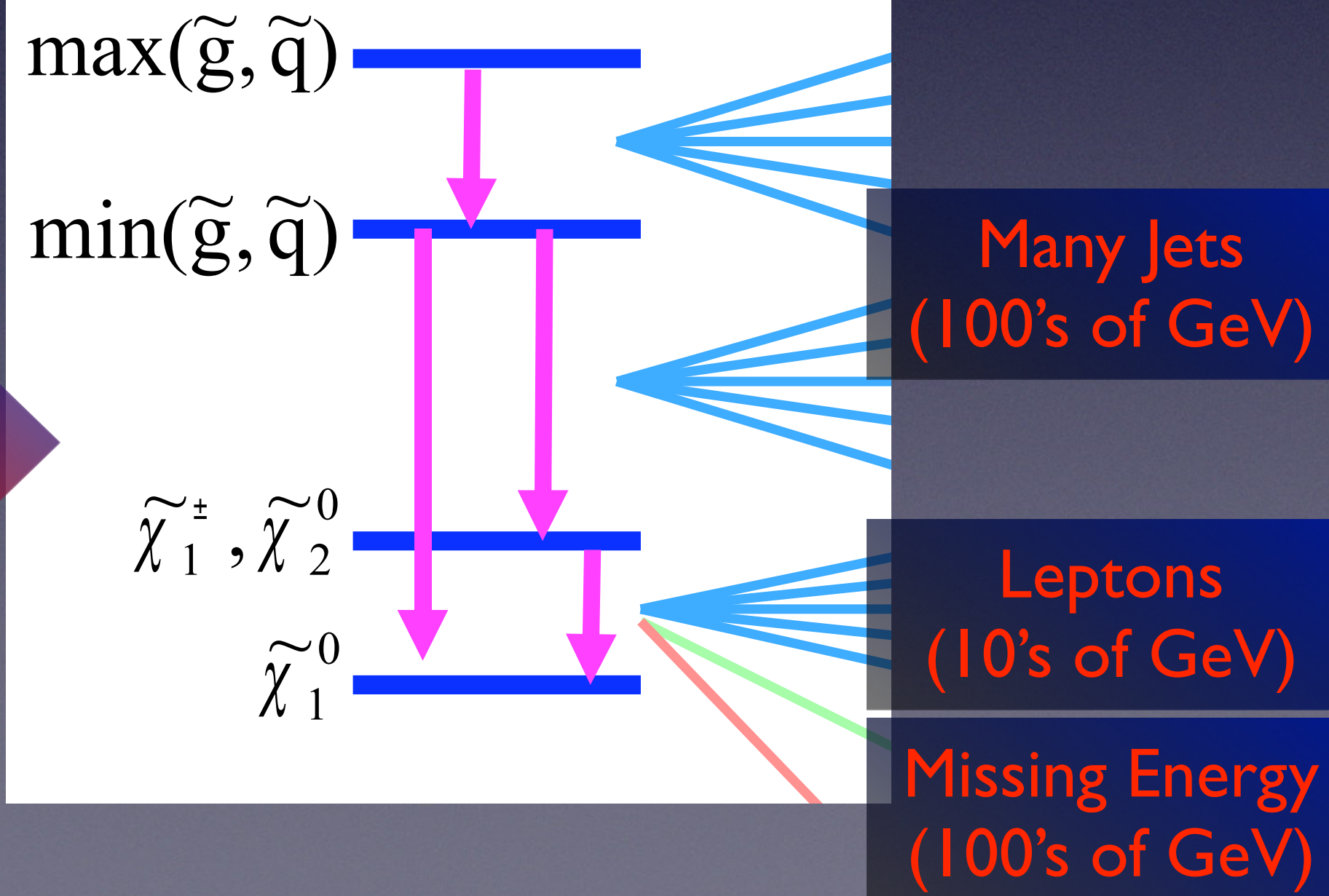
AMSB

$$\frac{H^\pm}{H^0 A^0} \quad \frac{\tilde{N}_4}{\tilde{N}_3} \quad \frac{\tilde{g}}{\tilde{C}_2} \quad \frac{\frac{\tilde{d}_L \tilde{u}_L}{\tilde{u}_R \tilde{d}_R}}{\frac{\tilde{b}_2}{\tilde{t}_2}} \quad \frac{\tilde{b}_1}{\tilde{t}_1}$$

mSUGRA

[illegible]

Gluinos and squarks
heavier than charginos/
neutralinos



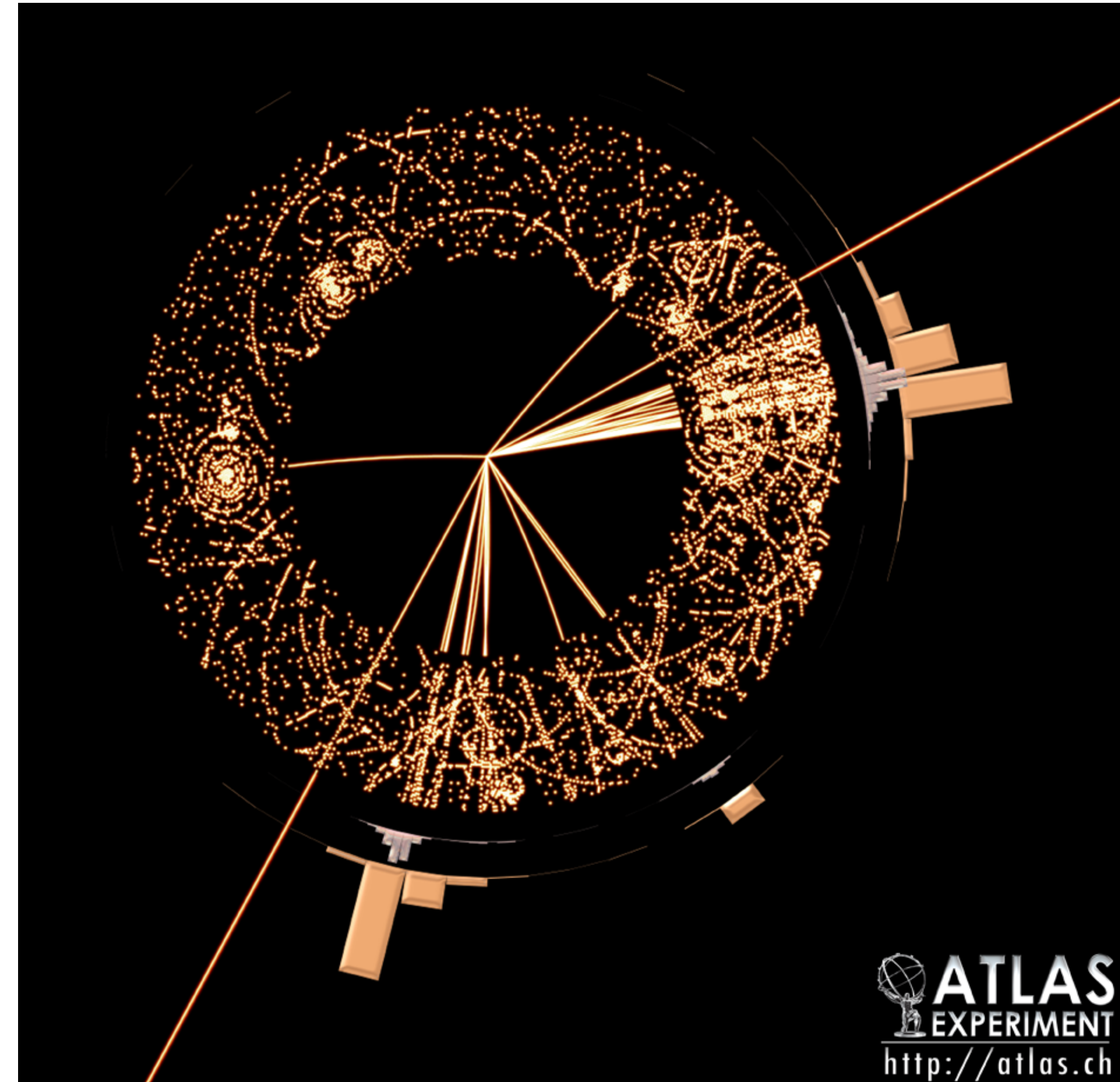
Inclusive Signatures

Signature	Motivating Model(s)	Comments
1 Jet + 0 Lepton + MET	• Large Extra Dim (ExoGraviton) • strong qG production, G propagate in extra Dim • Planck Scale is MD in $4+\delta$ dim • Normal Gravity $\gg R$ • SUSY • $qg \rightarrow \text{ISR} + 2 \text{ Neutralino}$ or squark + Neutralino	• Not primary discovery channel for SUGRA, GMSB, AMSB... but helps in characterization • Possible leading discovery for neutralino NLSP with nearly degenerate gluino
2,3,4 [b]-Jet + 0 Lepton + MET	• Squark/gluino production • squark $\rightarrow q + \text{LSP}$, gluino $\rightarrow q + \text{squark} + \text{LSP}$	• Possible leading squark/gluino discovery channel • Must manage QCD bkg
2,3,4 [b]-Jet + 1 Lepton + MET	• squark/gluino production with cascades which include electroweak (or partner) decays • high $\tan \beta$ leads to more b/t/τ's	• Lepton requirement suppresses QCD • τ's partially covered by e/μ
2 lepton + MET	• Same sign: gluino cascade can have either sign lepton... squark/gluino prod can produce same sign. • Opposite sign: squark/gluino decay mediated by Z (or partner) • Same flavor: 2 leptons from same sparticle cascade must be same flavor	• Reduced SM backgrounds for same sign • Opposite Sign-Flavor Subtraction
3 lepton + MET	• SUSY events ending in Chargino/neutralino pair decays • Weak Chargino/Neutralino production • Exotic sources	• Low SM bkg
2 photon + MET	• GMSB models with gravitino LSP and neutralino or stau NLSP • UED- each KK partons cascade to LKP which decays to graviton + γ	• No SUSY limit (not sensitive at the time)

SUSY/Higgs Data

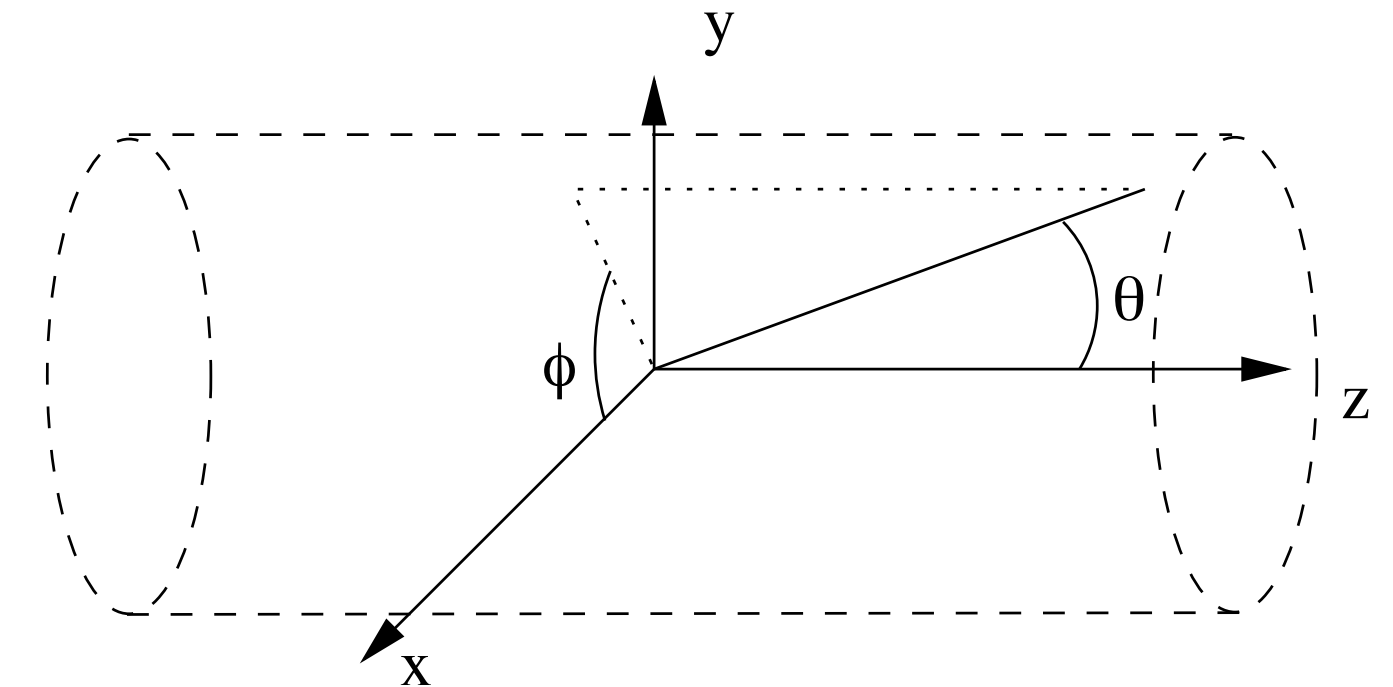
LHC Event

- The trigger LHC data is reconstructed into particle candidates.
 - Reconstruction algorithms try to find, identify, and measure particles
 - Typically there is one set of algorithms for each type of particle (corresponding to unique way the particle interacts with detector). For example:
 - *electron / photon*: Look tight deposits of energy in Electromagnetic Calorimeter. If a charged track points to it → electron otherwise photon.
 - *muon*: find track segment in inner detector and outer muon tracker.
 - *jets*: Look for broad deposits of energy in the Hadronic Calorimeter.
 - Not exclusive: Most electrons get also reconstructed as jet.
 - A process of Particle Identification and Overlap removal attempts to associate energy deposits in the detector with one unique hypothesis (as best as possible).
- Therefore the data and simulated data for an analysis will be process up to a level where every event consists of:
 - List of particle candidates, each with their momenta: electrons, photons, muons, electrons, jets (+ b-tagging), tau-jets, ...
 - Event level quantities: missing energy



LHC Events

- Candidate variables:
 - For each candidate, we measure it's 3-vector in a coordinate system best suited for Hadron Colliders.
 - Reflects the fact that not all of energy of beam goes into collision → can only apply energy conservation transverse to the beam.
 - $(p_x, p_y, p_z) \rightarrow (p_T, \eta, \phi)$
 - Once you choose what to call a particle, then you can assume a mass and turn your 3-vector → 4-vector
- Event Variables: Missing transverse energy (magnitude and direction in transverse plane (ϕ))
- In this context, by raw features we mean these 4-vector and event variables.



$$p_T^2 \equiv p_x^2 + p_y^2 \quad \eta \equiv \frac{1}{2} \ln \frac{1 + \cos \theta}{1 - \cos \theta}$$

$$\vec{p}_T \equiv \vec{p} \sin \theta \quad = -\ln \left(\tan \frac{\theta}{2} \right)$$

$$E_T^2 \equiv p_T^2 + m^2$$

$$= E^2 - p_z^2$$

Data Sample

- SUSY Data: (Selected events with only 2 leptons)
 - “signal”: The truth. 1 = signal. 0 = background.
 - 2 leptons: "l_1_pT", "l_1_eta", "l_1_phi", "l_2_pT", "l_2_eta", "l_2_phi"
 - Missing energy: "MET", "MET_phi"
 - Features: "MET_rel", "axial_MET", "M_R", "M_TR_2", "R", "MT2", "S_R", "M_Delta_R", "dPhi_r_b", "cos_theta_r1"
- Higgs Data: (Selected events with 1 lepton and 4 jets)
 - lepton pT, lepton eta, lepton phi
 - missing energy magnitude, missing energy phi,
 - jet 1 pt, jet 1 eta, jet 1 phi, jet 1 b-tag,
 - jet 2 pt, jet 2 eta, jet 2 phi, jet 2 b-tag,
 - jet 3 pt, jet 3 eta, jet 3 phi, jet 3 b-tag,
 - jet 4 pt, jet 4 eta, jet 4 phi, jet 4 b-tag,
 - Features: m_jj, m_jjj, m_lv, m_jlv, m_bb, m_wbb, m_wwbb.

High Level Features

- In this context, high-level features mean observables that are calculable from the 4-vectors.
 - Generally motivated by physics.
- For example: if you are expecting your signal and/or background events to include particles that are the decay products of a heavy particle,
 - you can attempt to identify the decay products and compute the mass of the heavy particle.
 - W bosons decays to 2 jets (m_jj), 3 jets (m_jjj), lepton+neutrino (m_lv), lepton+jet+neutrino (m_jlv), or 2 b-jets (m_bb)
 - Top quark decays into W + 2 b-jets (m_wbb)
 - These are an attempt to *exclusively* reconstruct W and top particles.
 - Exclusive: look for a specific particle by pull together *all* of the decay particles
- Another example: if you expect your signal and/or background events to consist of various types of particles with common traits,
 - you can attempt to construct features that capture the trait:
 - SUSY particles are heavy, therefore sum of the momenta of their decay products is large.
 - In SUSY, the last particle in the decay chain is not detected → large missing energy that isn't a fluctuation (e.g. same direction as a jet).
 - particles in SUSY events are more broadly distributed → “spherical” rather than “jetty” events.

$$H_T(n) \equiv \sum_{i=1}^n p_{ji,T} \quad M_{eff} \equiv H_T(4) + \sum_{k=1}^m p_{lk,T} + E_T^{miss}$$